Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	VeriGraphRAG: Verifiable and Reliable Enterprise Retrieval-Augmented Generation with Graph-Based Evidence and Risk Controls			
Korea's 12 National Strategic Technology Fields	Artificial Intelligence			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information and Communication		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Internet SW*	*Added Value/After-Sales Services*	*Culture, Medical, and Environmental based Knowledge Representation/ Intelligent Convergence Technologies*
Concept of Proposed Technology	Korean SMEs increasingly want "enterprise AI assistants" for engineering, customer support, compliance, and internal knowledge search. However, standard LLM chatbots can hallucinate, omit critical caveats, and provide answers without verifiable evidence. We propose **VeriGraphRAG**, a **verifiable and reliable RAG system** that (i) retrieves evidence from SME-owned documents/data, (ii) constructs an **evidence graph** (entities, relations, process links, citations), and (iii) generates answers with **verification and risk controls**, including: - **Evidence-grounded generation**: every key claim must link to retrieved passages/records (traceability). - **Consistency & contradiction checks**: detect conflicting sources and force			

	abstention/escalation. - **Policy-aware retrieval**: restrict what can be retrieved/shown based on roles and data sensitivity. - **Policy-aware retrieval**: restrict what can be retrieved/shown based on roles and data sensitivity. Potential uses: customer support copilots, internal engineering Q&A, QA/defect knowledge bases, regulatory/compliance document assistants, and technical sales enablement.
Rationale for Joint R&D Project	A research institute–SME collaboration is required because: 1. **Realistic enterprise data** (internal docs, manuals, tickets, ERP/CRM logs) is essential to design and validate trustworthy RAG, but it is only available within SMEs. 2. SMEs need **production constraints** addressed (latency, cost, access control, deployment), while RPI contributes novel algorithms for reliability, evaluation, and graph-based grounding. 3. Joint development accelerates **field validation** with measurable business impact (faster support resolution, reduced error rate, improved knowledge reuse), which is difficult to demonstrate with public benchmarks alone.
Key Words	Verifiable RAG Knowledge Graph-based grounding Safe and reliable AI Enterprise knowledge assistant Hallucination detection & provenance

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	TRL 3-4	End of the Project	TRL 7-8
R&D Objectives	□ Develop a **graph-structured evidence layer** for enterprise RAG that improves factuality and traceability. □ Develop **reliability mechanisms** (contradiction detection, abstention policies, risk scoring, audit trails). □ Build an **evaluation framework** for SME scenarios (accuracy, hallucination, provenance completeness, policy compliance). □ Deliver a **pilotable prototype** integrated with a Korean SME's document system/workflows.			
R&D Target Specifications	□ **Factuality / hallucination:** reduce unsupported claims by **≥50%** vs. baseline RAG on SME test set. □ **Provenance completeness: ≥90%** of non-trivial answers include **citation coverage** for all key claims (claim-to-evidence mapping). □ **Contradiction handling:** detect contradictory evidence with **F1 ≥ 0.80** on a curated contradiction set; trigger abstention/escalation correctly. □ **Latency (pilot setting):** average response time **≤ 3–5 seconds** for typical queries (with caching / adaptive retrieval). □ **Access control compliance: 0 critical violations** in red-team tests for restricted documents (role-based retrieval filters).			
R&D Plan	**Phase 1 (Months 0–3): Requirements + data onboarding** • Define SME use cases (e.g., support tickets, engineering manuals, compliance Q&A) • Build secure data pipeline + document ingestion; define access policies • Create evaluation protocol + initial test suite **Phase 2 (Months 4–9): Evidence graph + retrieval & grounding** • Design evidence graph schema (doc → passage → entity/process link) • Develop retrieval methods (hybrid retrieval + graph expansion) • Implement claim-to-evidence alignment **Phase 3 (Months 10–15): Reliability & safety layer** • Contradiction detection, abstention thresholds, escalation rules • Risk scoring (provenance completeness, conflict, uncertainty) • Monitoring dashboard (audit trail, provenance view) **Phase 4 (Months 16–24): Pilot prototype + validation** • Deploy pilot in SME environment (staging → limited production) • KPI evaluation (accuracy, resolution time, user satisfaction) • Finalize documentation + tech transfer package			

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	□ Lead algorithm R&D: graph-based grounding, verification, contradiction detection, evaluation methodology □ Build core prototype modules (retrieval, evidence graph builder, reliability scoring, monitoring) □ Design benchmarking suite and run controlled experiments □ Provide guidance on deployment architecture and model selection for efficiency
Expected R&R of Korean SME Partner	□ Provide target use case(s), domain requirements, and representative datasets/documents (with proper permissions) □ Define user roles and access-control policies; provide SME IT integration support □ Participate in iterative testing, user feedback sessions, and pilot validation □ Support commercialization planning and product integration roadmap

5. Expected Outcome/Impact

Expected Outcomes and Impact	**Technical outcomes:** • A verifiable enterprise RAG system with evidence graphs, reliability scoring, and contradiction-aware answering • An SME-focused benchmark and evaluation protocol for safe/reliable enterprise assistants **Commercialization potential:** • SME can package this as a differentiated enterprise product feature (trust, auditability, policy compliance) • Potential for deployment across multiple SME verticals (manufacturing, logistics, energy, medical devices, SaaS) **Broader impact:** • Reduces operational errors caused by hallucinations in enterprise copilots • Improves knowledge reuse and workforce productivity while maintaining accountability and traceability

6. References

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☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization